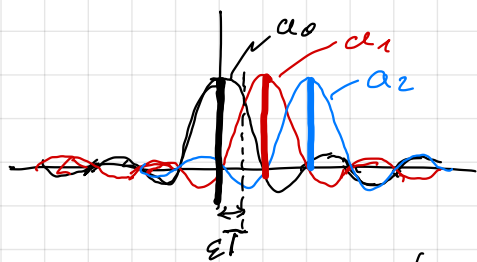


Wireless Receivers

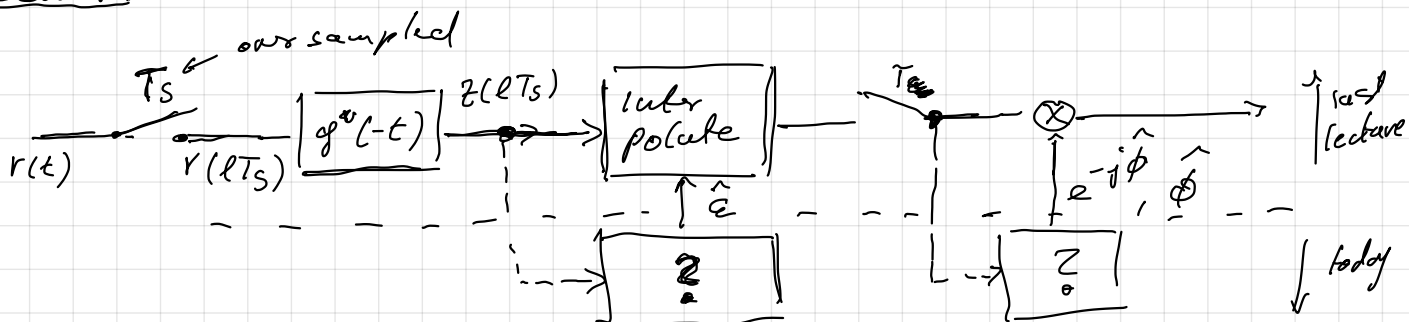
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Parameter Estimation



Sampling-time offset ϵT_s

Receiver:



How to get to $\hat{\epsilon}$ and $\hat{\phi}$?

Receiver Parameter Estimation

Formulate the ML condition

We have: $\underline{r} = [\dots, r(-2T_s), r(-T_s), r(0), r(T_s), r(2T_s), \dots]$ (oursampled samples)

Unknowns: transmitted symbols $\underline{a} = [\dots, a[-2], a[-1], a[0], a[1], a[2], \dots]$

Time offset: ϵ

Phase offset: ϕ

$$\{\hat{\underline{a}}, \hat{\underline{\epsilon}}, \hat{\phi}\} = \arg \max_{\underline{a}, \underline{\epsilon}, \phi} Pr(\underline{r} | \underline{a}, \epsilon, \phi)$$

Blind estimation: data is unknown

a) Joint estimation: search over all options

b) Estimate parameters one-by-one

→ marginalize over the others = take the average over the others

Timing $\hat{\epsilon}$: $Pr(\underline{r} | \epsilon) = \int \sum_{\underline{a}} Pr(\underline{a}) Pr(\underline{r} | \epsilon, \underline{a}, \phi) p(\phi) d\phi$

uniform

uniform $-\pi \dots +\pi$

- Phase $Pr(\underline{r}|\phi) = \int \sum_{\underline{a}} Pr(\underline{a}) Pr(\underline{r}|\phi, \underline{a}) p(\underline{a}) d\underline{a}$

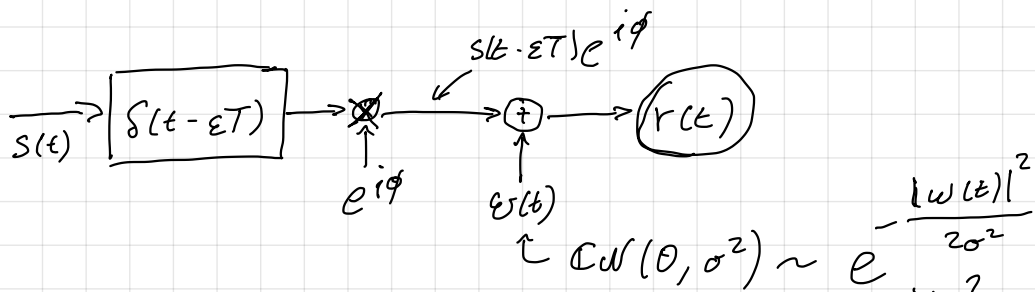
3) Estimation one after another
 $\rightarrow$ for each parameter assume we know the ones we estimated before perfectly

- Timing $\underline{\epsilon}$: $\hat{\underline{\epsilon}} = \arg \max_{\underline{\epsilon}} Pr(\underline{r}|\underline{\epsilon})$

- Phase ϕ : $\hat{\phi} = \arg \max_{\phi} Pr(\underline{r}|\underline{\epsilon}=\hat{\underline{\epsilon}}, \phi)$

- Data $\underline{a}$: $\hat{\underline{a}} = \arg \max_{\underline{a}} Pr(\underline{r}|\underline{\epsilon}=\hat{\underline{\epsilon}}, \phi=\hat{\phi}, \underline{a})$

ML criterion for AWGN channel $\rightarrow$ Goal is to define $Pr(\underline{r} | \dots)$



$$\underline{r}(t): \mathcal{CN}(\underbrace{s(t - \epsilon T)e^{j\phi}}_{\text{Tx Sequence incl. phase offset}}, \sigma^2) \sim e^{-\frac{|r(t) - s(t - \epsilon T)e^{j\phi}|^2}{2\sigma^2}}$$

Ref RX signal (no noise) + given ϵ, ϕ
 $s(t - \epsilon T)e^{j\phi}$

Instead of comparing to the delayed Ref sequence $s(t - \epsilon T)e^{j\phi}$ we compare $s(t)e^{j\phi}$ to an "undelayed" RX sequence/samples $r(t + \epsilon T)$ and write $\underline{r}(t + \epsilon T)$ as a function of $\underline{r}(t)$ [$\underline{r}(tT_s)$]

Consider $p_r(\underline{r} | \underline{r}^{\text{ref}})$

$$\text{For one sample: } p(r | r^{\text{ref}}) = \exp(-|r - r^{\text{ref}}|^2) \quad \sum_n |r_n - r_n^{\text{ref}}|^2$$

$$\begin{aligned} \text{for multiple samples: } p(\underline{r} | \underline{r}^{\text{ref}}) &= \prod_n p(r_n | r_n^{\text{ref}}) = \exp(-\|\underline{r} - \underline{r}^{\text{ref}}\|^2) \\ &= \exp(-\cancel{\|\underline{r}\|^2} + 2\text{Re}\{(\underline{r}^{\text{ref}})^H \underline{r}\} - \|\underline{r}^{\text{ref}}\|^2) \end{aligned}$$

Note: $\|\underline{r}\|^2$ is indep. of the parameters we try

has no impact on finding the maximizing parameters

$$\mathcal{L}(\underline{r} | \underline{r}^{\text{ref}}) \sim \exp(2\text{Re}\{(\underline{r}^{\text{ref}})^H \underline{r}\} - \|\underline{r}^{\text{ref}}\|^2) \quad r(t + \epsilon T) \leftrightarrow$$

Compare the "corrected" sampled sequence $r(t + \epsilon T)$ with sampled TX output (w/o time delay)

$$r(t) \rightarrow \underline{r}(t + \epsilon T) = \underline{z}_e(\epsilon) \quad \text{with} \quad \underline{r}^{\text{ref}} \rightarrow s(t) = \sum_k a_k g(t - kT) e^{j\phi} \Big|_{t = \epsilon T} = \underline{a}_e e^{j\phi}$$

$$\mathcal{L}(\underline{z}_e(\epsilon) | \underline{a}_e e^{j\phi}) = \exp(+2\text{Re}\{\underline{a}_e^H e^{-j\phi} \underline{z}_e\} - \|\underline{a}_e\|^2)$$

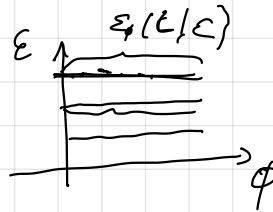
$$\mathcal{L}(\underline{z} | \underline{a} e^{j\phi}) = \exp(+2\text{Re}\{\underline{a}^H e^{-j\phi} \underline{z}\} - \|\underline{a}\|^2)$$

$$= \exp \left(\sum_{l=-L}^{+L} \left[2 \operatorname{Re} \{ \alpha_l^* z_l(\varepsilon) e^{-j\phi} \} \right] - \sum_{l=-L}^{+L} |\alpha_l|^2 \right)$$

for large L
 $\rightarrow$ const.

$$= \exp \left(\sum_{l=-L}^{+L} 2 \operatorname{Re} \{ \alpha_l^* z_l(\varepsilon) e^{-j\phi} \} \right)$$

derived from our N
 samples @ $r(LT_s)$



Blind estimation average over all possible $\underline{a}$, or α_l $l=-L \dots L$

$$\mathbb{E}_{\underline{a}} \{ L(\underline{z} | \underline{a} e^{j\phi}) \} \rightarrow \underline{a} \text{ is in the exponent}$$

$$\underline{\mathbb{E}}_{\underline{a}} \{ f(\underline{a}) \} \neq f(\underline{\mathbb{E}}_{\underline{a}} \{ \underline{a} \})$$



Workaround: Taylor expansion

$$\exp \left(2 \operatorname{Re} \{ \sum \alpha_l^* z_l(\varepsilon) e^{-j\phi} \} \right) = 1 + 2 \operatorname{Re} \{ \sum \alpha_l^* z_l(\varepsilon) e^{-j\phi} \} + \left(\operatorname{Re} \{ \sum \alpha_l^* z_l(\varepsilon) e^{-j\phi} \} \right)^2$$

$$\textcircled{1} \mathbb{E}_{\underline{a}, \phi} \{ 2 \operatorname{Re} \{ \sum \alpha_l^* z_l(\varepsilon) e^{-j\phi} \} \} = \operatorname{Re} \{ \mathbb{E}_{\underline{a}, \phi} \{ \sum \alpha_l^* z_l(\varepsilon) e^{-j\phi} \} \}$$

$$\textcircled{2} = \left(\frac{1}{2} \sum_l \alpha_l^* z_l(\varepsilon) e^{-j\phi} + \frac{1}{2} \sum_l \alpha_l z_l^*(\varepsilon) e^{+j\phi} \right)^2 \leftarrow \text{remove the } \operatorname{Re} \{ \dots \}$$

$$= \sum_l \left(|\alpha_l|^2 |z_l(\varepsilon)|^2 + \frac{1}{2} \operatorname{Re} \{ \alpha_l^2 (z_l(\varepsilon))^2 e^{j2\phi} \} \right)$$

Average $\underline{\mathbb{E}}_{\underline{a}}$

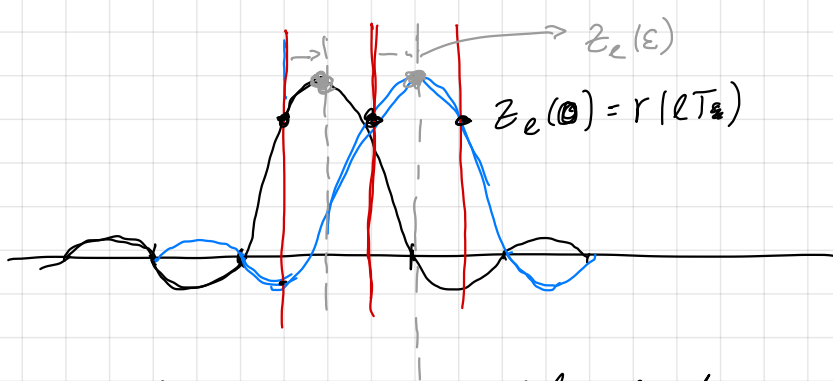
$$= \mathbb{E}_{\phi} \left\{ \sum_l \underbrace{\mathbb{E}_{\underline{a}} \{ |\alpha_l|^2 \}}_{\rightarrow \text{const.}} |z_l(\varepsilon)|^2 + \frac{1}{2} \operatorname{Re} \left\{ \underbrace{\mathbb{E}_{\underline{a}} \{ \alpha_l^2 \}}_{\text{const.}} (z_l(\varepsilon))^2 e^{j2\phi} \right\} \right\}$$

Average over $\phi \in [0, 2\pi]$ $\rightarrow$ eliminates the 2nd term

$$L(\varepsilon) \sim \sum |z_l(\varepsilon)|^2 \quad \text{with } z_l(\varepsilon) = r(LT_s + \varepsilon T)$$

ε write as a fct.
 of $r(LT_s)$

$$\hat{\varepsilon} = \arg \max_{\varepsilon} \sum |z_l(\varepsilon)|^2$$



$$\underline{L(\epsilon)} \sim \sum \underline{|z_e(\epsilon)|^2}$$

with $z_e(\epsilon) = r(lT_s + \epsilon)$ ✗

we only have $r(lT_s)$

write $z_e(\epsilon)$ as a fct. of the samples I have $r(lT_s)$

How to construct $z_e(\epsilon)$ from $r(lT_s)$

Interpolate: band-limited signal

1) write as Fourier series, evaluate at $lT + \epsilon$

$$\underline{|z_e(\epsilon)|^2} = \sum_{n=-\infty}^{\infty} \underbrace{c_n}_{\text{offset}} \underbrace{e^{j2\pi n \epsilon}}_{\text{Fourier coeffs centered around the lth symbol}}$$

2) Get $c_n^{(l)}$ from our samples $r(pT_s)$ around $r(lT)$

the symbol l $lT, lT+T_s, lT+2T_s, lT+3T_s, \dots, lT+(M-1)T_s$

M -fold oversampling

$$c_n^{(l)} = \sum_{p=0}^{M-1} |r(lT + pT_s)|^2 e^{-j \frac{2\pi}{M} np}$$

3) Substitute into $L(\epsilon) \sim \sum_{\epsilon} |z_e(\epsilon)|^2 = \sum_{n=-\infty}^{\infty} \underbrace{\sum_l}_{\text{over all samples}} \underbrace{\sum_{p=0}^{M-1}}_{\text{over all samples}} |r(lT + pT_s)|^2 e^{-j \frac{2\pi}{M} np} e^{j2\pi n \epsilon}$

Note: $r(t)$ is band limited to $\pm \frac{1}{2T}$

$\Rightarrow |r(t)|^2$ is bandlimited too, but to $\frac{1}{T}$

$\rightarrow$ only c_{-1}, c_1 and $c_0 \neq 0$

$$\underline{L(\epsilon)} \sim c_0 + \underbrace{c_1 e^{-j2\pi \epsilon} + c_{-1} e^{j2\pi \epsilon}}_{\text{conjug. conj pairs}} = c_0 + 2 \operatorname{Re} \left\{ \underbrace{c_1 e^{-j2\pi \epsilon}}_{\text{must}} \right\}$$

$$c_1 = \sum_{\epsilon} \sum_p |r(lT + pT_s)|^2 e^{-j \frac{2\pi}{M} p} = \text{FT of squared rx samples}$$

$$\underline{\epsilon} = \arg(c_1) \frac{1}{2\pi}$$

should be real for max

Phase Estimation $\rightarrow$ Assume that $\varepsilon = \hat{\varepsilon}$
 Timing is already corrected
Data aided: know data $\underline{a}^{(0)}$ known

$$L(\phi) = L(\underline{z} | \underline{a} = \underline{a}^{(0)}, \varepsilon = \hat{\varepsilon}, \phi) = \exp \left(\operatorname{Re} \left\{ \sum_l \underline{a}_l^{(0)*} z_l(\hat{\varepsilon}) e^{j\phi} \right\} \right) \rightarrow \max$$

$\rightarrow$ Real valued

$$\hat{\phi} = \arg \left(\sum_l \underline{a}_l^{(0)*} z_l(\hat{\varepsilon}) \right) \quad \checkmark \quad \text{Ambiguity of } \pm \pi$$

Non-data aided: not knowing $\underline{a}$

Problem: $L(\phi)$ depends on $\underline{a}$ or a_l

PSK:

Idea: Remove dependency on $a_l \leftarrow$ single symbol

- Assumption: PSK-modulation $z_l(\hat{\varepsilon}) = a_l e^{j\phi}$ $a_l = \underline{c}$
- Note: $(a_l)^Q = \left(e^{j \frac{2\pi}{Q} d_l} \right)^Q = e^{j 2\pi d_l} \quad d_l = \text{int.}$
 $\equiv 1$ regardless of d_l

Consider modified RX-signal $\underline{[z_l(\hat{\varepsilon})]}^Q = \underline{a_l^Q} e^{jQ\phi}$

$$L(\phi) = \operatorname{Re} \left\{ \sum_l \underline{(\underline{a}_l^{(0)})^Q} \underline{[z_l(\hat{\varepsilon})]}^Q e^{-jQ\phi} \right\}$$

$$\hat{\phi} = \arg \left(\sum_l \underline{[z_l(\hat{\varepsilon})]}^Q \right) \frac{1}{Q} \quad \text{Ambiguity of } \pm \frac{\pi}{Q}$$

$\pm \pi$

